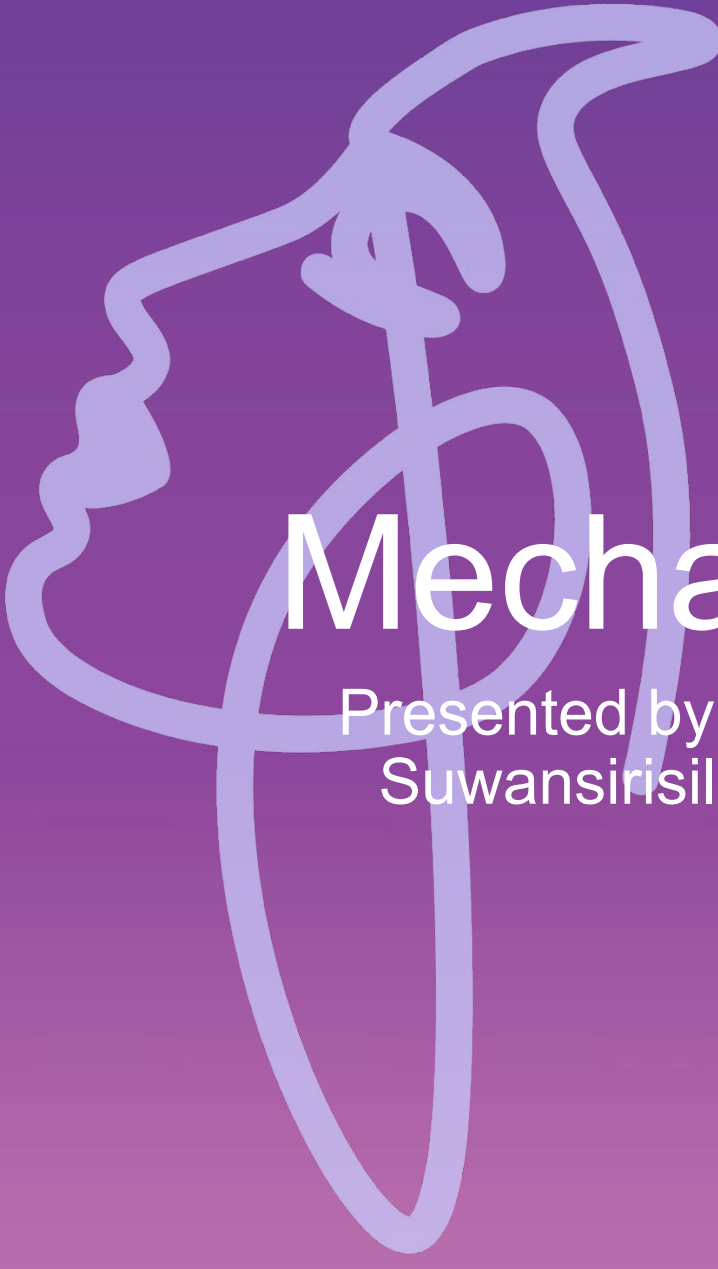




Mechanical Team Proposal

Presented by: Grace Kilfoy, Clare Pastula, Sadie Williams, Jina Suwansirisilp, Tomiris Akhmedina, Pepper Hedquist, Linh Le

Goals & Requirements

Problem Definition

Initially designed for 500 scmh, but with a projected increase to 800 scmh by 2030, the new compressor system aims to increase its production of RNG.

Goals:

- Solution should use safety precautions & ensure control of transient conditions such as load changes
- Prototype should
 - respond to fault scenarios without causing unstable pressure spikes
 - minimize gas loss by maintaining continuous stable flow
 - clearly show flow control, pressure response, and system interaction

Constraints:

- Solution must
 - maintain 130 psi discharge pressure
 - prevent over-pressurization of the recycle and buffer tanks (max. 7psi)
 - operate under an increased landfill gas input of up to 800 scmh
- Prototype must
 - Be completed in 4-6 weeks
 - simplified enough to be converted to CAD modelling and simulation

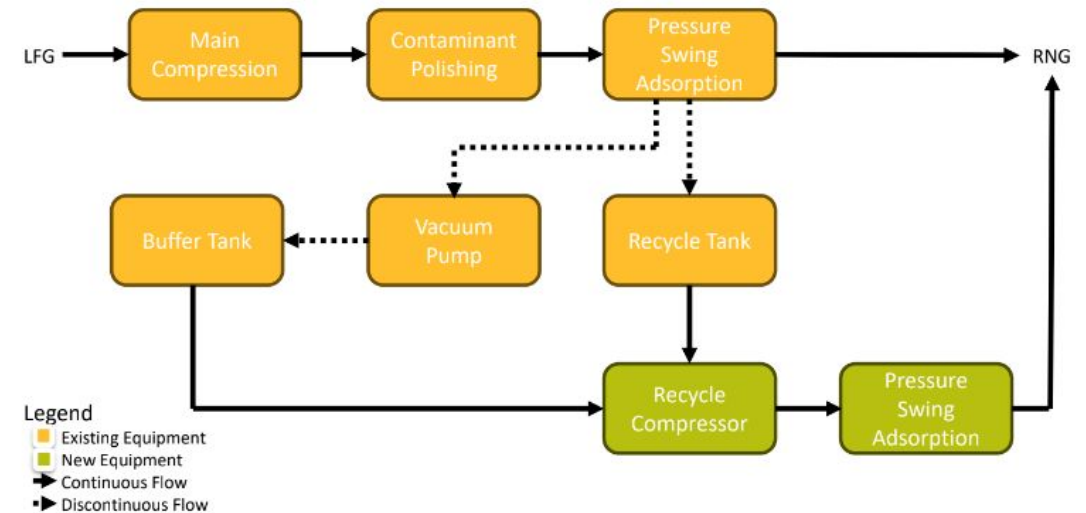
Solution

Proposed Solution

We propose the installation of a second 100 HP recycle compressor in parallel with the existing unit.

Evaluation Criteria:

- Technical Feasibility
- Operational Reliability
- Cost-Effectiveness
- Safety



Evaluation of Solution

Technical Feasibility

- Parallel compressor matches existings LeROI unit for compatibility
- Enables continuous operation without surge issues
- Allows maintenance isolation without stopping RNG production

Parameter	Value
Motor size	100 HP
Capacity	713 m ³ /h per unit
Control strategy	VFD with PID pressure control
Installation factor	4.87 × equipment cost

Evaluation of Solution

Operational Reliability

- **Before upgrade:** Frequent venting events, pressure excursions, risk of compressor trip
- **After upgrade:** No venting required, redundant system enables maintenance without shutdown, reduced load per compressor

3. Flow rate during blowdown

$$\text{Flow rate (m}^3/\text{h)} = \frac{V_{\text{gas released (m}^3\text{)}}}{T_{\text{blowdown (s)}}} \cdot 3600$$

1. $\frac{29.92}{60} \cdot 3600 = 1796 \text{ m}^3/\text{h}$

2. $\frac{7.51}{20} \cdot 3600 = 1352 \text{ m}^3/\text{h}$

Rotary. $\frac{5.87}{10} \cdot 3600 = 2113 \text{ m}^3/\text{h}$

⇒ All three > compressor rate = 713 m³/h

Flow rate during blowdown of current compressor calculation

Evaluation of Solution

Safety Assessment

Provides safety through

- built -in backup
- reduced operational stress on each individual compressor

Risk	Before Upgrade	After Upgrade
Over-pressurization (recycle tank >7 psi)	High - frequent venting events	Low - adequate compression capacity
High discharge temperature (>230°F)	Critical - operating near limit	Reduced - load shared between units
Gas venting (methane release)	1.8 million m ³ /year	None
Single point failure	Compressor failure = system shutdown	The second compressor allows continued operation

Evaluation of Solution

Economic Evaluation

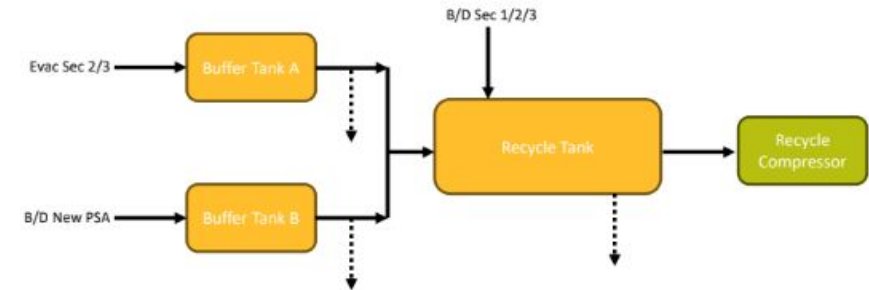
- Total cost (capital + operating)
= \$998,394 ~ \$1,001,734
- Pays itself back within 11 months of operation

Metric	Value
Capital investment	\$965,446
Net annual savings	\$1,036,942 - \$1,045,282
Payback period	~11 months

Prototype

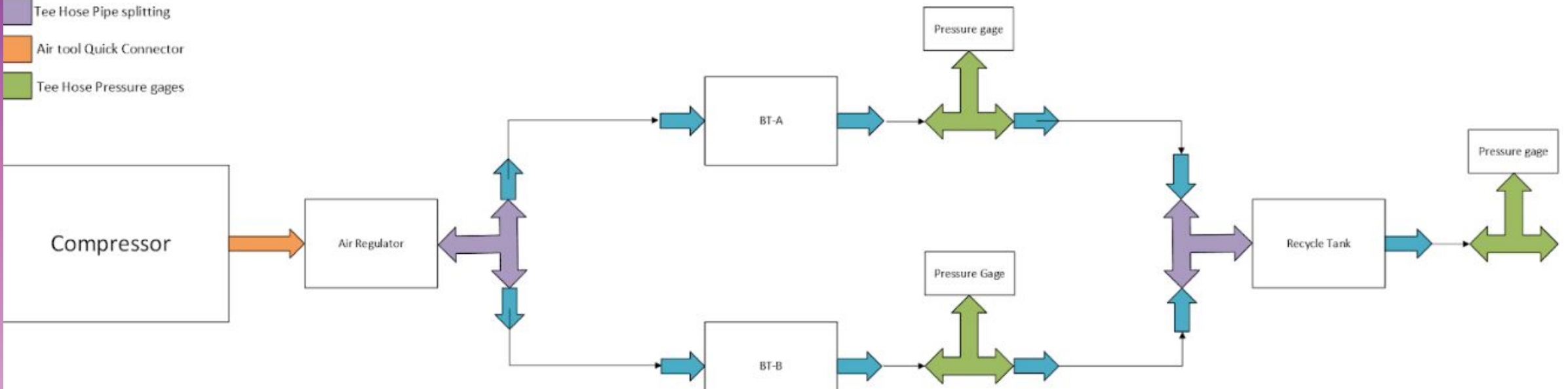
Design

The New Plant Operation

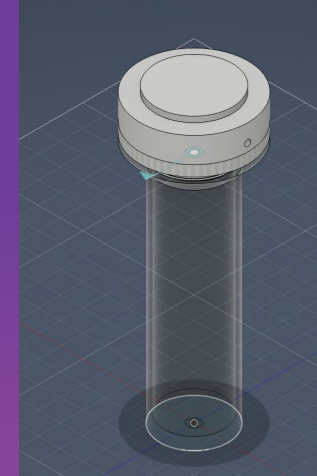
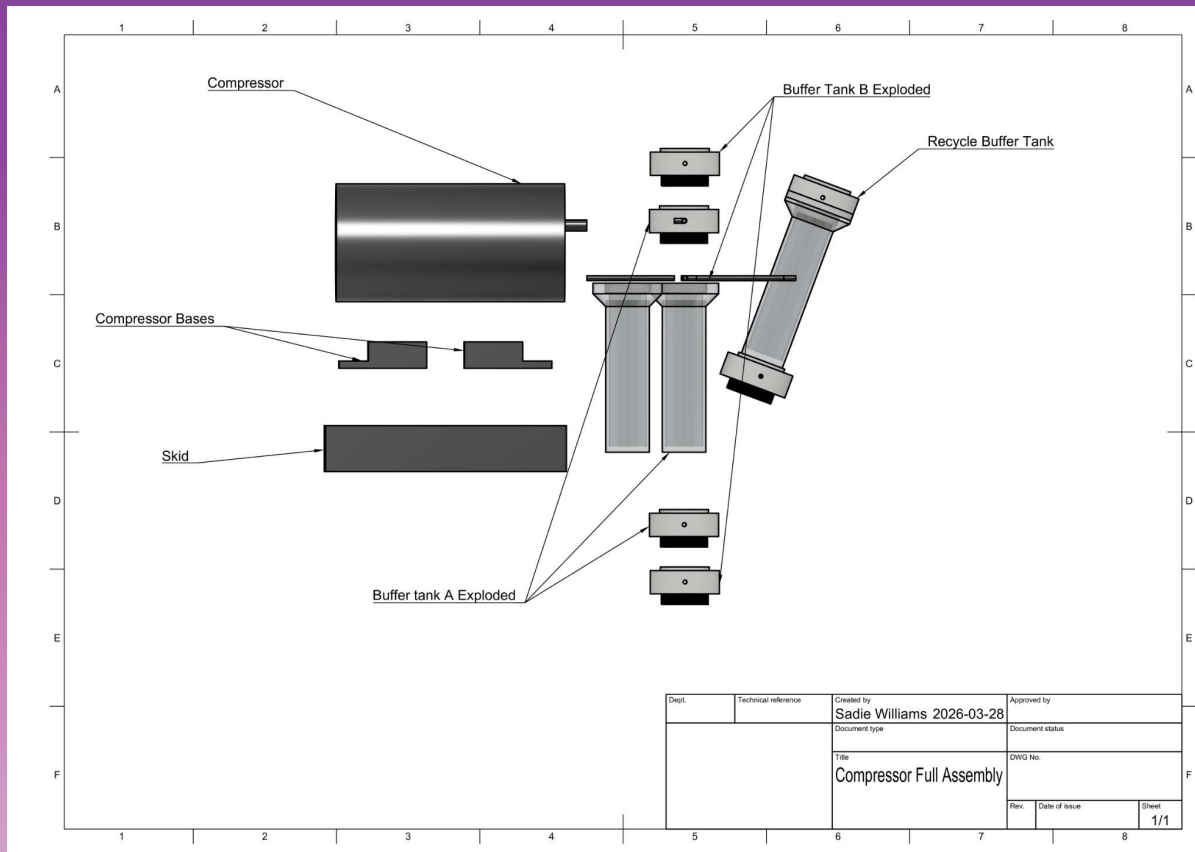


Legend
■ Buffer Tank
■ Compressor
→ Normal Flow
- - - Purge Lines

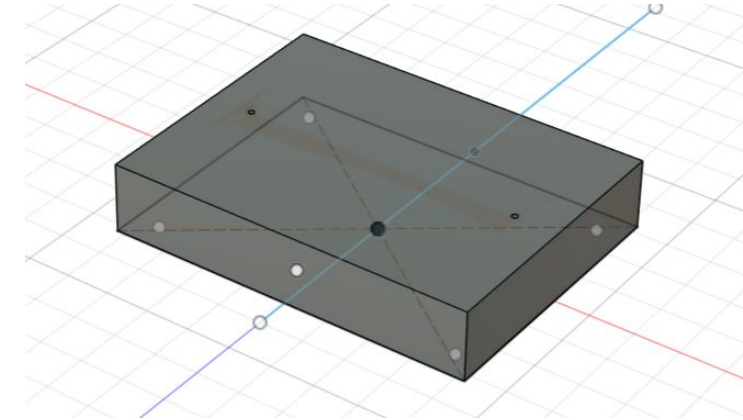
- Barb-NPT Connector
- Tee Hose Pipe splitting
- Air tool Quick Connector
- Tee Hose Pressure gages



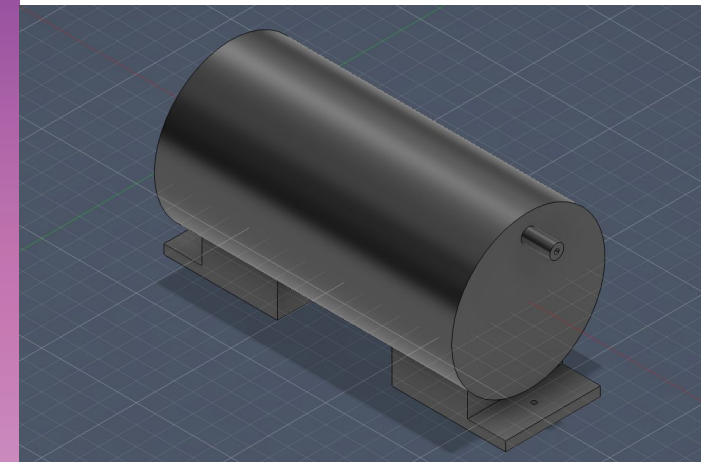
CAD Models



Buffer tank/ recycle tank



Compressor mounting plate



Compressor

Physical Model

Approach

- Combined hands-on prototyping with analytical evaluation
- Help visualizing the compressor operations within the system
 - Gas flow
 - Recycling behaviours
 - Pressure changes
- Air compressor acts as a parallel unit in the prototype system
- Designed to closely resemble the compressor system at KBGP



Testing Procedure

To minimize leaking from the prototype:

- Sealed all valves
- Ensured equal pressurization between pressure gages
- Utilized soap and water to find leaks

For Safety:

- Utilized both control valves
- Implemented maximum pressure of 20 PSI

Video Demonstration

Implementation

Theoretical Implementation

- Add a parallel compressor to the existing Kelowna Biogas recycle compressor system
- Integrate with current equipment to improve gas recovery

Practical Implementation

- Researched FortisBC requirements and compressor options
- Developed theoretical system flowcharts and design plans
- Modelled compressor parts and assembly in Autodesk Fusion
- Procured materials, assembled physical prototype and tested
- Documented all design and build stages
- Managed progress with Gantt Chart

Engineering Challenges

Organizational:

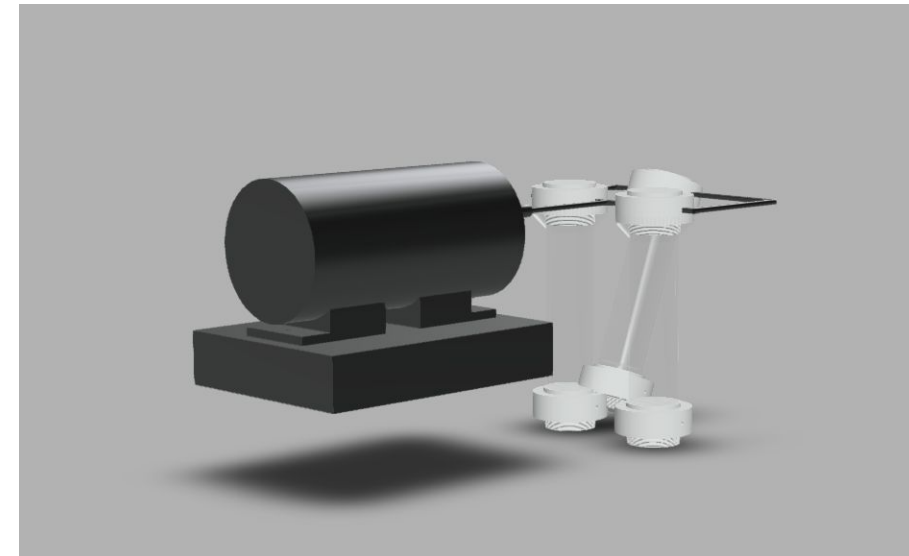
- Time Constraints
- Team Availability

Design:

- CAD Software
- Multiple designers

Build:

- Improper Components
- Procurement Lead Times



Tasks	Team Member	Week of Feb 23				
		M	T	W	R	F
Planning	All					
Evaluate Design Solutions: New Parallel vs Compressor Upgrade	LK					
Updated System Flowchart	CP					
Calculate annual cost of gas lost	LK					
Research & Documentation	SW					
CAD (imperial units) ** tubing connections 1/4"	All					
Buffer Tanks >No THREAD	PH					
Compressor Skid/Mounting Interface	CP					
Pump & Motor (motor & pump one piece)	TA, G					
Recycle tank	JS					
Compressor	GK					
Inlet Pipe	GK					
Assembly	CP & SW					
Simulation						
CFD Simulation	G					
Build						
Physical Prototype BoM	All					
Construction	All					
Testing	All					
Team Presentation	All					
Slides Preparation	All					
Presentation (TBD)	All					
Team Report	All					

Conclusion

Conclusion

Through utilization of calculations, research, CAD modelling and prototype development, the mechanical team has determined that installing a parallel compressor is the most effective solution.

This Solution:

- Allows better blowdown gas management
- Maintains required discharge pressure of 130 Psi
- Reduces gas loss caused by pressure surges
- Meets the project goals by improving
 - reliability
 - methane recovery
 - supporting projected plant growth

Thank You!
Questions?

Appendix

Bill of Materials

Mechanical System	Quantity	Supplier/Manufacturer	Unit Price	Total Price
Piping	2	Home Depot	\$7.20 / 10 ft	\$14.40
Compressor	1	Home Depot	\$149.00	\$149.00
Compressor Battery/Charger	1	Home Depot	\$128.00	\$128.00
Air regulator	1	Home Depot	\$37.98	\$37.98
1/4" M Connector (comp-regulator)	2	Home Depot	\$8.98	\$17.96
Barb-NPT Connectors (piping to tanks)	4	Amazon	\$10.99 / 4	\$43.96
Hose Clamps	2	Amazon	\$9.99 / 10	\$19.98
Thread Tape (for sealing)	1	Home Depot	\$2.22	\$2.22
Buffer tank/recycle tanks *	3	Amazon	\$56.29	\$168.87
1/4" Tee Fitting *pressure gauge connection	2	Amazon	\$16.49	\$32.98
1/4" Tee Fitting (relief Valve) *split hoses	2	Home Depot	\$9.48	\$18.96
Pressure Gages	4	Amazon	\$16.97	\$67.88
Pressure Gages	3	Home Depot	\$11.31	\$33.93
Valves	3	Home Depot	\$17.98	\$53.94
Male Air Tool Quick Connector	1	Canadian Tire	\$5.99	\$5.99

\$796.05

Gas Loss Calculation

PSA - Section 1			PSA - Section 2/3			PSA - Rotary (New)		
Item	Value	Unit	Item	Value	Unit	Item	Value	Unit
Online Pressure	120	psig	Online Pressure	110	psig	Online Pressure	130	psig
Online Time	300	sec	Online Time	180	sec	Online Time	120	sec
Equalization Time	50	sec	Equalization Time	50	sec	Equalization Time	15	psig
Blowdown Pressure	40	psig	Blowdown Pressure	45	psig	Blowdown Pressure	2	psig
Blowdown Time	60	sec	Evacuation Pressure	2	psig	Evacuation Pressure	0.75	m3
Blowdown Valve Diameter	4	inch	Evacuation Time	240	sec	Empty Volume	10	sec
Evacuation Pressure	5	psig	Empty Volume	1.7	m3	Blowdown Time	20	sec
Evacuation Time	500	sec	Blowdown Time	20	sec	Blowdown Valve Diameter	2	inch
Empty Volume	5.5	m3	Blowdown Valve Diameter	2	inch			

Online Pressure	The pressure when the vessel is in adsorption mode (making BKG)	- KBP has 4 parallel PSA trains
Online Time	How long it stays in adsorption before regenerating	- Each has 3 adsorption vessels
Blowdown Pressure	The pressure it drops TO during blowdown	- At any given time: 4 in adsorption, 8 in regeneration
Blowdown Time	How long the blowdown takes	
Empty Volume	The physical size of the vessel (how much gas it holds)	

PSA Section	Start Pressure (psig)	Start Pressure (psia)	End Pressure (psig)	End Pressure (psia)	Vessel Volume (m³)
Section 1	120	134.7	40	54.7	5.5
Section 2/3	110	124.7	45	59.7	1.7
Rotary (New)	130	144.7	15	29.7	0.75

1. Convert unit (psig → psia (absolute))

2. Gas released / blow down

Gas volume (m³) = Vessel volume (m³) × $\frac{P_{start} - P_{end}}{P_{atm}}$

Section 1:

$$V_{gas/blowdown} = 5.5 \times \frac{134.7 - 54.7}{14.7} = 29.98 \text{ m}^3$$

Section 2:

$$V = 1.7 \times \frac{124.7 - 59.7}{14.7} = 7.51 \text{ m}^3$$

Section 3:

$$V = 0.75 \times \frac{144.7 - 29.7}{14.7} = 5.87 \text{ m}^3$$

3. Flow rate during blowdown

Flow rate (m³/h) = $\frac{V_{gas released (m³)}}{T_{blowdown (s)}} \times 3600$

1. $\frac{29.98}{60} \times 3600 = 1795 \text{ m}^3/\text{h}$

2. $\frac{7.51}{20} \times 3600 = 1352 \text{ m}^3/\text{h}$

Rotary: $\frac{5.87}{10} \times 3600 = 2113 \text{ m}^3/\text{h}$

→ All three > compressor rate = 713 m³/h

4. Gas lost / event

$V_{gas lost} = \frac{(Blowdown Flow rate (m³/h) - Comp)}{3600}$

1. $V = \frac{(1795 - 713)(60)}{3600} = 18.05 \text{ m}^3$

2. $V = \frac{(1352 - 713)(20)}{3600} = 3.55 \text{ m}^3$

Rotary: $V = \frac{(2113 - 713)(10)}{3600} = 3.89 \text{ m}^3$

5. Frequency of blowdown

Assuming 1 event / cycle

of blowdown / day

1. Total cycle time / vessel: 500 + 50 + 60 + 500 = 910s

blowdown / day = $\frac{24 \times 3600}{910} = 95$

2. Total cycle time / vessel: 180 + 50 + 20 + 140 = 490s

blowdown / day = $\frac{24 \times 3600}{490} = 176$

Rotary: Total cycle time / vessel: 120 + 10 = 130s

blowdown / day = $\frac{24 \times 3600}{130} = 664$

blowdown / year

1. 95 × 365 = 34675

2. 176 × 365 = 64240

Rotary: 664 × 365 = 242360

6. Total annual gas lost

1. 18.05 × 34675 = 625883.75 m³/year

2. 3.55 × 64240 = 228052 m³/year

3. 3.89 × 242360 = 942780.4 m³/year

7. Annual cost of gas lost

Total = 1796716.15 m³/year

1796716.15 × 0.2 = \$1038029.69 / year

